Xiao Fu

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Education

Zhejiang University Hangzhou, Zhejiang
Bachelor in Information Engineering | College of Information Science & Electronic Engineering *Sept. 2018 – Jul. 2022*

- **GPA:** 3.96/4.00, 90.54/100, **Ranking:** 4/145
- **National Scholarship awardee, Twice**
- **Chu Kochen Honors College:** Intensive Training Program of Innovation and Entrepreneurship (ITP) (Honored minor for the selected top 40 students with innovation capability from 6,000 freshmen)
- **Courses:** Computer Vision(96),Digital Signal Processing(95),Matrix Theory(94),Stochastic Process(96),Probability and Mathematical Statistics(95),Numerical Analysis(92),Signals and Systems(98),Calculus(A)(90)

Publications

$*$ indicates equal contribution

- [1] Guangyi Zhang*, **Xiao Fu***, Qiyu Hu, Yunlong Cai, Guanding Yu, "Hybrid Precoding Design Based on Dual-Layer Deep-Unfolding Neural Network," *IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Helsinki, Finland, 2021, pp. 678-683 **[Paper][Code][Slides]**

Research Experience

Faster Semantic/Instance Labeling with Panoptic NeRF Max Planck Institute for Intelligent Systems
Supervisor: Dr. Yiyi Liao and Prof. Andreas Geiger *Jul. 2021 – Now*

- Propose Panoptic NeRF, which transforms coarse 3D bounding box annotations (static objects in outdoor KITTI360 dataset) into dense and consistent 2D pixel-wise and 3D point-wise semantic and instance annotations
- Panoptic NeRF, which is based on NeRF, takes DeepLab's pretrained results as 2D priors to overcome the ambiguity of labeling in 3D overlapping areas and utilizes LiDAR, fisheye, and stereo cameras to improve the NeRF's density
- Intend to publish an ECCV 2022 paper as the first author

Faster Solution of Optimization Algorithm with DLDUNN Zhejiang University
Supervisor: Prof. Guanding Yu *Jun. 2020 – Feb. 2021*

- Proposed a novel framework, i.e. a Dual-Layer Deep-Unfolding Neural Network(DLDUNN) which unfolds the iterative Penality Dual Decomposition (PDD) algorithm into a layer-wise structure whereby some trainable parameters are introduced into the places of the high-complexity operations, e.g. large-dimensional matrix inversions and iterative sub algorithms, so as to address the high computational complexity problem of the dual-layer PDD algorithm and presents the major original performance in the meantime
- **Laureate of Excellent Award(Team leader) - Student Research Training Program(Province-level)**

Honors and Scholarships [Certificate & Report]

- National Scholarship **(Highest Honor for Chinese Undergraduates, Twice)** *2020,2021*
- ISEE Honor Student Award **(Top 5 students among ISEE 314 undergraduates)** *2021*
- ISEE Yuelun Alumni Scholarship **(Top 10 teams out of ISEE undergraduates and graduates)** *2020,2021*
- First-Class Scholarship for Academic Excellence of Zhejiang Univeristy**(Top 3%)** *2020,2021*
- Government Scholarship for Academic Excellence of Zhejiang Province **(Top 3%)** *2019*
- Finalist Winner – Mathematical Contest in Modeling (MCM/ICM) **(Top 2% out of 20,954 teams)** *2020*
- Meritorious Winner – Mathematical Contest in Modeling (MCM/ICM) **(Top 7% out of 26,112 teams)** *2021*
- Gold Prize – 7^{th} International "Internet+" Innovation and Entrepreneurship Contest **(Team Leader)** *2021*
- Gold Prize – 12^{th} "Challenge Cup" National College Student Business Plan Competition *2020*
- Grand Prize – "Yongdian Cup" Innovation and Entrepreneurship Contest **(Team Leader)** *2019*
- Grand Prize – "Midea" Global Technology Innovation Contest *2020*
- Academic Excellence Scholarship of Zhejiang University *2019,2020,2021*
- Zhejiang University Five-Star Volunteer *2021*

浙江大学 信息与电子工程学院
College of Information Science & Electronic Engineering, Zhejiang University

CERTIFICATE

This is to certify that Mr Xiao Fu ranks the 4th among the 145 students with the specialty of Information Engineering.

College of Information Science & Electronic Engineering

Zhejiang University

28/09, 2021

浙江大学 信息与电子工程学院

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